Edward Zhang

zhange2@sas.upenn.edu | (484) 743-7678 | LinkedIn | GitHub | Website

EDUCATION

University of Pennsylvania | *B.S.E. Computer Science and Mathematics* | *GPA: 3.93/4.0* **Expected Graduation: May 2028**
Relevant Coursework: Data Structures and Algorithms, Advanced Algorithms, Computer Systems, Discrete Mathematics, Probability, Advanced Linear Algebra, Big Data Analytics, Theory of Computation, Automata and Complexity, Computer Organization and Design

EXPERIENCE

Carrier | Software Engineer Intern – Beverly, MA **June 2026 – Aug 2026**
- Built backend infrastructure for a release intelligence platform automating validation package generation across 7 product projects.
- Designed async ingestion workflows to normalize 100+ Jira issues, validation fields, test cycles, and executions into PostgreSQL.
- Developed containerized FastAPI/Celery services across PostgreSQL, Redis, MinIO/S3, AWS ECS, Podman, and CircleCI.
- Reduced manual validation-package preparation time by 70% by automating Jira/Zephyr metadata ingestion and synchronization.

Hong Kong Science & Technology Parks | Software Engineer Intern – Hong Kong **Jun 2025 – Aug 2025**
- Deployed a fault-tolerant distributed system on Kubernetes, handling peak traffic of 2,000+ daily transactions with auto-scaling.
- Engineered an asynchronous job processing engine using Redis queues and health checks with 99.9% uptime during traffic spikes.
- Optimized backend infrastructure and resource allocation strategies, reducing cloud compute costs by 25% through request batching.
- Developed a high-throughput pipeline automating the digitization of 10,000+ documents, with OCR microservices.

Hack4Impact | Project Manager/Technical Lead – Philadelphia, PA **Sep 2024 – Present**
- Led development of a full-stack web platform (React, Node.js, PostgreSQL), serving 1.5M+ users across 5,000+ schools.
- Implemented REST APIs and normalized database schemas, improving query latency by 65% in production load.
- Coordinated a ~10-person team through design reviews, API standardization, and code reviews to unblock backend development.
- Introduced code quality and reliability practices, including structured code reviews, API versioning, and backend testing workflows.

Rajapakse's Lab | Research Software Developer – Philadelphia, PA **Jan 2025 – Aug 2025**
- Designed and implemented a scalable preprocessing and feature extraction pipeline for 3D CT imaging data (50K+ DICOM files).
- Optimized volumetric data ingestion by batching and layout-aware preprocessing, reducing experiment iteration time by 25%.

Wharton Analytics Fellows | Senior Data Analyst – Philadelphia, PA **Aug 2025 – Present**
- Collaborated with Accenture to implement a parameterized cost estimation model, analyzing 30+ business functions and 32 agents.
- Engineered data pipelines aggregating public and enterprise data sources to support model inputs and calibration.
- Developed interactive dashboards to visualize cost structures and tradeoffs of AI deployment strategies across large organizations.

Discrete Mathematics Teaching Assistant (CIS 1600) – Philadelphia, PA **May 2025 – Present**
- Led recitations, office hours, and review sessions for 250+ students covering proofs, combinatorics, and graph theory.

PROJECTS

Distributed Task Queue System | Go, Redis, Kubernetes
• Built a distributed task queue with worker orchestration, retry policies, dead-letter queues, and fault-tolerant async execution.
• Implemented Redis-backed scheduling, batching, and worker heartbeat mechanisms to coordinate jobs across Kubernetes pods.

World Cup Prediction Platform | React, FastAPI, PostgreSQL, Redis, Docker
• Built a live tournament analytics platform ingesting match data to update standings, advancement odds, and knockout scenarios.
• Designed REST APIs, normalized PostgreSQL schemas, and Redis-cached background jobs for low-latency prediction updates.

AWARDS & PROGRAMS

USACO Gold Division (Top 1,000 competitors globally), PennApps XXIV Best Community Impact Hack, CalHacks 12.0 Snap Spectacles Track 2nd Place, PwC Engineer Your Career Program, Hong Kong Olympiad in Informatics Bronze Medallist & Training Team (Top 20 competitors in HK), Hong Kong Secondary School Coding Challenge Bronze Medallist, UK Mathematics Challenge Gold Award x5

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, OCaml, Go | **Web & APIs:** React, TypeScript, Node.js, REST, GraphQL, Pydantic, FastAPI
Systems & Data: SQL, PostgreSQL, Redis, MongoDB, Kafka, Elasticsearch, Docker, Kubernetes, AWS, Azure, Linux, AI tools (Claude)